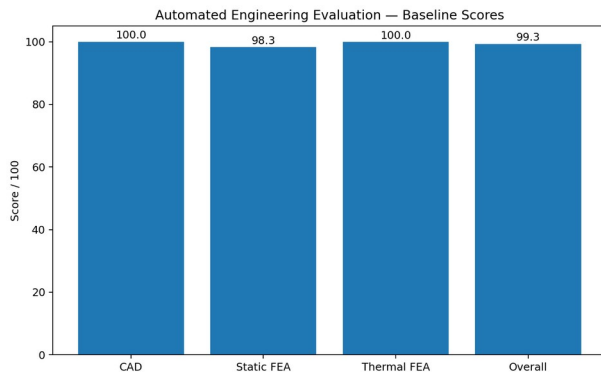


AMR IMAD AWAD

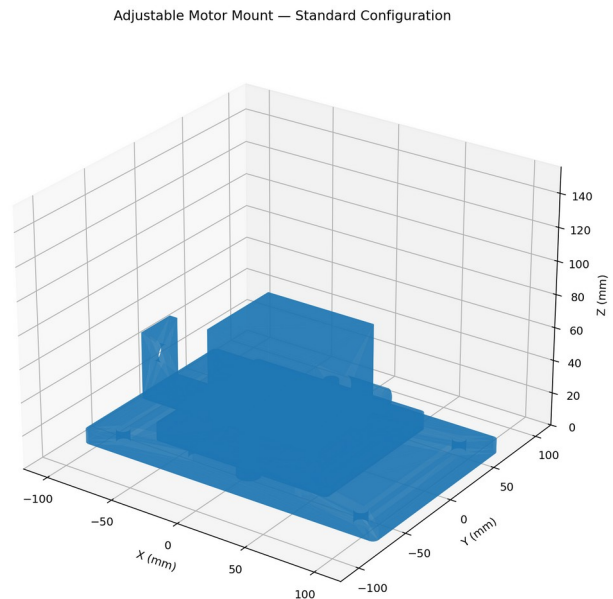
MECHANICAL ENGINEERING PORTFOLIO

Parametric CAD | FEA Validation | Thermal Analysis | Python Engineering Evaluation

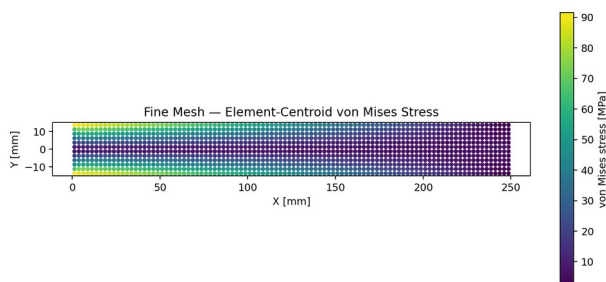
Palestine | +972 59 580 4921 | amrawad0595@gmail.com | [LinkedIn](#)



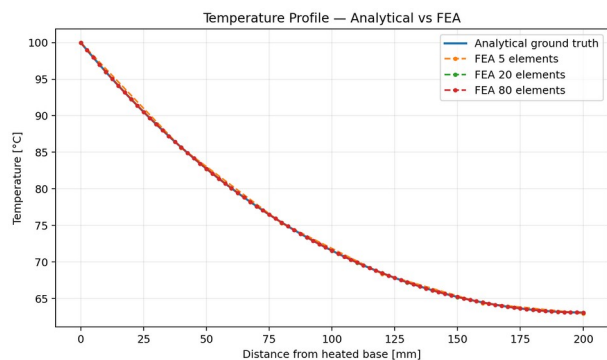
Automated evaluation



Parametric CAD



Static FEA



Thermal validation

SELECTED WORK FOR CAD EVALUATION / MECHANICAL ENGINEERING ROLES

Five focused projects demonstrating a repeatable workflow: build the geometry, define the physics, establish ground truth, validate the numerical result, and automate the evaluation.

Engineering work built around evidence, not screenshots.

Mechanical Engineering graduate (2026) with project work spanning parametric CAD, structural and thermal validation, Python-based evaluation, and mechanical mechanism design. This portfolio is intentionally selective: each project shows the engineering problem, the implementation, the validation method, and the measurable result.

5	3	99.3/100	2026
selected projects	validation domains	automated baseline	B.Sc. Mechanical Eng.

CORE CAPABILITIES

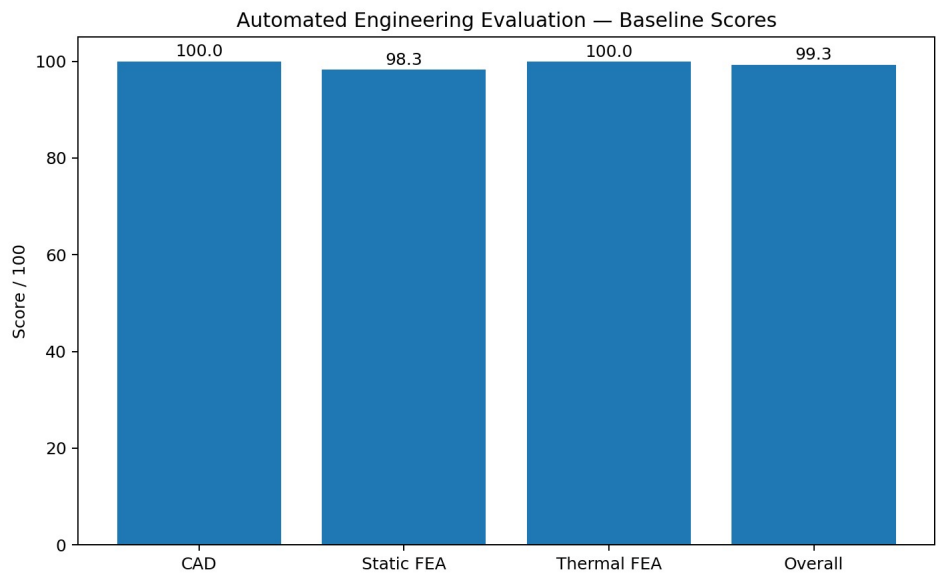
CAD & DESIGN	ANALYSIS & VALIDATION
• SolidWorks and CadQuery • Parametric geometry and configurations • STEP / STL / DXF workflows • Mechanical assemblies and design intent • BOM and manufacturing documentation	• Static structural FEA • Stress / deformation interpretation • Thermal conduction and convection • Mesh convergence and ground truth • Python deterministic evaluation

SELECTED PROJECTS

PROJECT	ENGINEERING QUESTION	EVIDENCE	TOOLS
Automated Evaluation Harness	Can engineering outputs be graded consistently?	PASS/FAIL + weighted scoring	Python, JSON, CSV
Parametric CAD Benchmark	Does geometry remain valid across configurations?	27 deterministic checks	CadQuery, STEP, DXF
Static FEA Benchmark	Does FEA reproduce known mechanics?	0.52% deflection error	FEA, beam theory
Thermal FEA Benchmark	Does numerical heat transfer match ground truth?	0.0012% tip-temp error	Thermal FEA, fin theory
Compact Scissor Lift	Can a mechanism be taken from requirements to design package?	CAD + kinematics + BOM	CAD, calculations, drawings

Automated Engineering Evaluation Harness

Python | deterministic scoring | CAD + static FEA + thermal FEA

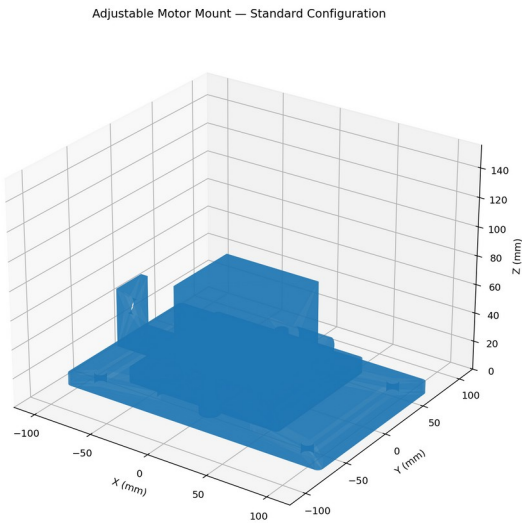


Baseline engineering scores generated by the evaluation harness.

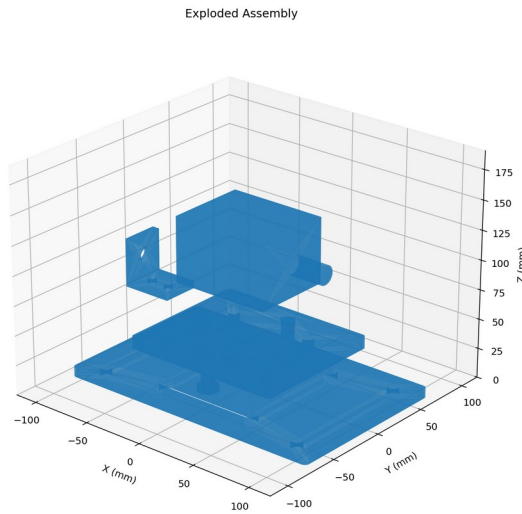
99.3/100		27/27	44.818/100	3/3
overall baseline		CAD checks passed	injected failure set	unit tests passed
Challenge		Engineering outputs can look plausible even when geometry, constraints, boundary conditions, convergence, or energy balance are wrong.		
Scope		Built a configurable Python harness that ingests CAD, static FEA, and thermal-validation results and evaluates them against explicit engineering criteria.		
Method		Threshold checks, weighted scoring, critical-failure rules, SHA-256 input fingerprints, machine-readable reporting, and repeatability tests.		
Validation		Known-good inputs must pass, corrupted inputs must fail, and identical inputs must return identical reports.		
Result		Baseline: PASS at 99.3/100. Injected failure set: FAIL at 44.818/100. Unit tests confirmed baseline acceptance, failure rejection, and deterministic output.		
Evidence		Python source, rules JSON, baseline/failure inputs, reports, score summaries, and unit tests.		

Parametric CAD Validation Benchmark

Adjustable motor mount | configurations | deterministic geometry checks



Standard configuration

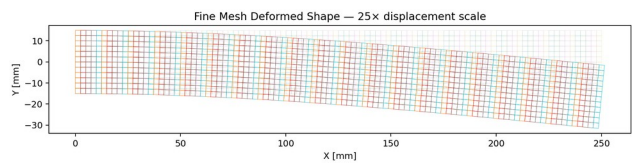


Exploded assembly

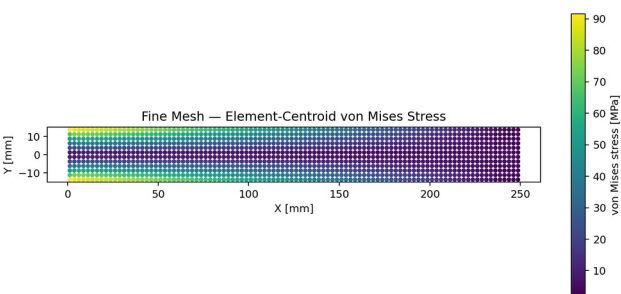
3	27/27	5	+40 mm
design configurations	validation checks	intentional failure cases	standard travel
Challenge		A parametric model should remain valid as dimensions change, while invalid parameter sets should be detected before downstream use.	
Scope		Created an adjustable motor-mount assembly with three configurations, controlled slider travel, neutral CAD exports, and automatic design-rule checks.	
Method		Validated clearances, component envelopes, travel, hole-pattern boundaries, and positive geometry build volume.	
Debug cases		Injected five invalid states: undersized slot, excessive travel, oversized slider, out-of-bounds motor pattern, and guide-hole interference.	
Result		All 27 baseline checks passed across three configurations; the validator identified the intentionally invalid parameter states.	
Deliverables		CadQuery source, STEP/STL/DXF, JSON/CSV validation reports, debug-case inputs, and SolidWorks import instructions.	

Static FEA Ground-Truth Benchmark

Cantilever plate | analytical mechanics | mesh convergence | equilibrium



Deformed shape, 25x visual scale

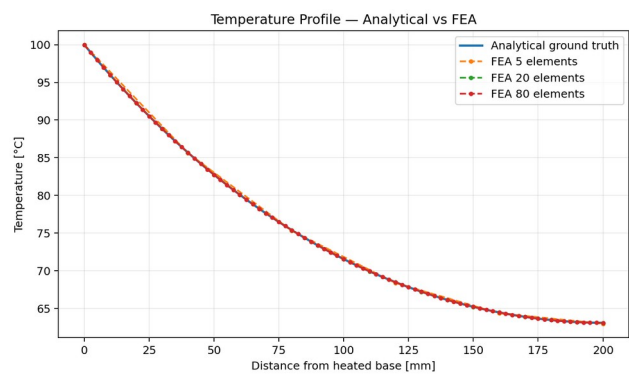


Element-centroid von Mises stress

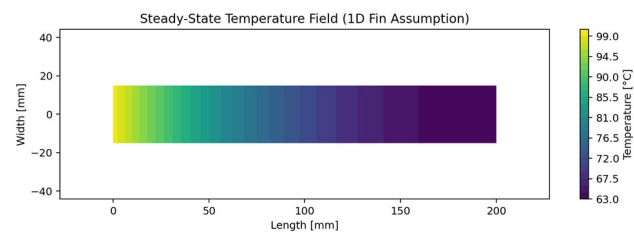
0.524%	0.335%	2.50	1200
tip-deflection error	stress error	nominal FOS	fine-mesh elements
Challenge		A solver result is only useful if loads, constraints, mesh, and physics are trustworthy.	
Model		250 x 30 mm cantilever, 10 mm thickness, 600 N end load; analytical root stress = 100 MPa and analytical tip displacement = 0.661376 mm.	
Method		Solved the same problem using a Q4 plane-stress finite-element implementation; evaluated coarse/medium/fine meshes and checked force and moment equilibrium.	
Validation		Fine-mesh tip displacement = 0.664841 mm (0.524% error). Defined interior stress sample differed by 0.335% from the analytical value.	
Equilibrium		Support reactions reproduced the applied 600 N load and 150,000 N-mm bending moment to numerical precision.	
Engineering takeaway		Compared physically equivalent quantities and avoided treating local fixed-edge stress peaks as direct beam-theory ground truth.	

Thermal FEA Ground-Truth Benchmark

Aluminum cooling fin | conduction + convection | energy balance



Analytical vs FEA temperature profile

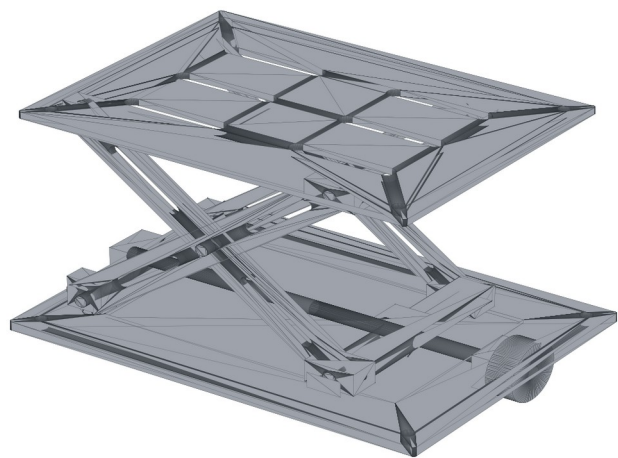


Steady-state temperature field

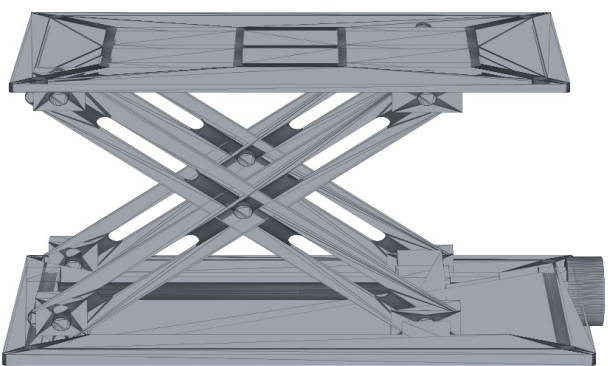
0.00119%		0.00149%		0.0%		80	
tip-temp error		heat-flow error		energy-balance error		fine-mesh elements	
Challenge				Thermal simulation should be checked against known physics rather than trusted because the contour plot looks reasonable.			
Model				200 x 30 x 4 mm aluminum fin; base at 100 C; ambient at 25 C; convection coefficient 15 W/m2-K.			
Method				Derived the exact convective-fin solution, built a finite-element conduction model with distributed convection, and performed mesh convergence from 5 to 80 elements.			
Validation				Fine-mesh tip temperature = 63.08147 C versus analytical 63.08192 C. Base heat input matched analytical heat flow within 0.00149%.			
Energy balance				Numerical base heat input and total convective heat loss closed with effectively zero reported energy-balance error.			
Deliverables				STEP/STL/DXF geometry, analytical derivation, temperature profiles, numerical solver, convergence results, ANSYS replication guide, and validation report.			

Compact Scissor Lift Mechanism

Self-directed mechanical design | kinematics | load path | design package



Isometric CAD model



Side-view mechanism geometry

10 kg	20-60 deg	~136 mm	TR12x3
rated centered payload	operating angle	linkage travel	lead-screw concept
Challenge	Develop a compact manual lifting mechanism from requirements through concept-level mechanical design documentation.		
Scope	Defined design requirements; developed CAD assembly and part models; performed kinematic and global analytical load calculations; sized the lead-screw concept; and prepared BOM, profiles, drawings, and a concept-stage risk review.		
Geometry	Base 320 x 240 x 8 mm; top platform 300 x 220 x 6 mm; scissor arm pivot spacing 260 mm; operating angle 20 to 60 degrees.		
Analysis	Evaluated mechanism height, actuation force, arm loads, buckling, pin bearing, lead-screw force/torque, and self-locking behavior.		
Status	Virtual concept design. CAD, calculations, BOM, drawings, and risk review complete. Physical prototype, local FEA, proof-load testing, and certification were not performed.		
Deliverables	STEP/STL assembly and parts, engineering calculation workbook, BOM, manufacturing drawings/DXF profiles, validation plots, test-plan document, and design report.		

How I approach engineering problems

The common thread across these projects is validation: define the requirement, model the system, establish a reference or acceptance criterion, test the output, and document what the result means. My goal is to build engineering work that is inspectable and reproducible - not just visually complete.

1	2	3	4	5
DEFINE	MODEL	REFERENCE	VALIDATE	REPORT
Requirements / assumptions	CAD / physics	Ground truth / limits	Errors / convergence	Evidence / decisions

ENGINEERING TOOLKIT

DESIGN & CAD

- SolidWorks
- CadQuery
- AutoCAD
- Fusion 360
- STEP / STL / DXF
- Technical drawings

ANALYSIS & DATA

- FEA fundamentals
- Static structural analysis
- Thermal analysis
- Python
- MATLAB
- Carrier HAP
- CSV / JSON workflows

ADDITIONAL WORK

Additional project work includes an industrial sheet-metal control enclosure with bend calculations, flat patterns, BOM, manufacturing drawings, and STEP/DXF deliverables; an EV battery-pack CAD assembly; and a Python-based innovative HVAC graduation project.

AMR IMAD AWAD

Mechanical Engineering Graduate | CAD, FEA & Engineering Evaluation

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Detailed project files, source code, CAD exports, calculations, and validation reports available with the portfolio package.